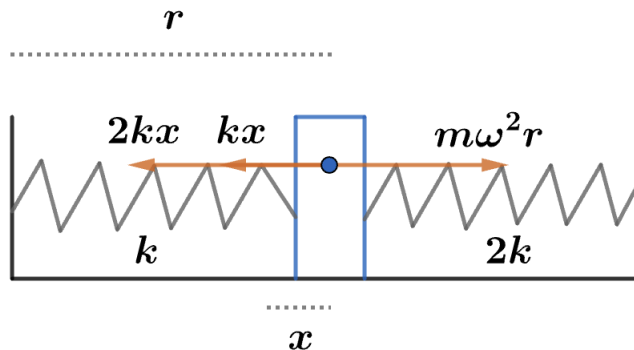


2024 F=ma Exam: Problem 10

Kevin S. Huang



We go to the rotating frame and add the (fictitious) centrifugal force $m\omega^2 r$ directed outward. Since the block is at rest, we balance forces

$$kx + 2kx = m\omega^2 r$$

where x is the displacement of the block. Since the block starts at distance l from the axis, we have

$$x = r - l$$

Solving for r ,

$$3k(r - l) = m\omega^2 r$$

$$(3k - m\omega^2)r = 3kl$$

$$r = \frac{3kl}{3k - m\omega^2}$$

so the answer is D.